

Negating statements; -

R

$\sim R$

$\Rightarrow$ Recall

De Morgan's law

$$\sim (P \wedge Q) = \sim P \vee \sim Q$$

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ex:-

R :- You can solve it by factoring or
with the quadratic formula.

$$\sim R: \sim (P \vee Q)$$
$$\sim P \wedge \sim Q$$

$$R := P \vee Q$$

ex:-

R :- The number x and y are both odd

$$\sim R := \sim (P \wedge Q)$$
$$\sim P \vee \sim Q \quad \begin{matrix} \top & \sim & \top \\ & \top & \end{matrix}$$

$$R := P \wedge Q$$

~~$\sim R$~~ $\sim R$:- neither x nor y are
odd $= (\sim P \wedge \sim Q)$

P :- x is odd

Q :- y is odd

$\sim R$:- ^{at least} one of x or y are even

$\sim R$:- x is even or y is even